Lecture 7

# Application I: molecular vibrations

This lecture counts the vibrational frequencies of the ammonia molecule and finds which of them are degenerate. It needs not a single force constant. We describe a molecule by the symmetry group of its equilibrium shape, its point group. The small oscillations about equilibrium are governed by the dynamical matrix, and the point group acts on the displacements of the atoms. The dynamical matrix commutes with that action, so Lecture 6's block theorem applies: the vibrational frequencies fall into families labelled by irreducible representations, and characters count them. The lecture analyses the vibrations of ammonia in full, and the tutorial those of water.

By the end of this lecture you will be able to:

- Recognise the point groups $C_{nv}$ and $D_{nh}$ from a molecule's shape.
- Compute the displacement character and remove the rigid motions.
- Classify the vibrations of a molecule by symmetry.

## 7.1 Molecules and their point groups

So far we have applied symmetry operations to triangles and polygons. Now we apply them to a real physical object — a molecule. No prior theory of molecules is needed: all we shall use is where the nuclei sit in space.

Picture a molecule as a few points in space. Each point is the nucleus of one atom at its equilibrium position, labelled by its species: the chemical element and, where it matters, the isotope. The water molecule $H_2O$, for instance, is three points: one oxygen and two hydrogens (Figure 7.1).

A *symmetry operation* of such a set of points is a rigid motion of space after which the molecule looks exactly the same. That means two things: the set of points is carried onto itself, and each nucleus is replaced by a nucleus of the same species. An oxygen nucleus may not land where a hydrogen was, and neither may a deuterium. Isotopes of one element feel the same forces but have different masses, and the masses enter the vibrations. The tutorial of Lecture 8 meets a molecule where this matters.

**Definition 7.1** (Point group)**.** The *point group* of a molecule is the group of all its symmetry operations under composition. The motions allowed are rotations, reflections and their combinations about the molecule's centre of mass, that is, the orthogonal transformations.

The group is called a point group because all these operations keep one point fixed, the centre of mass. We take that point as the origin.

Chemists and physicists write these operations in Schoenflies notation:

- $E$ — the identity, that is, "do nothing";
- $C_n$ — a rotation through $2\pi/n$ about some axis (the same symbol as the cyclic group this rotation generates);
- $\sigma$ — a reflection in a plane.

The rotation axis of highest order is called the *principal* axis. A reflection plane gets the subscript $v$ (*vertical*) if the principal axis lies in that plane. Planes that the group carries into one another share a name; planes that it does not are told apart by primes, $\sigma_v$ and $\sigma_v'$. The three vertical planes of ammonia, which its rotations carry into one another, are all $\sigma_v$.

The name of a point group is built from the same symbols. For instance $C_{2v}$ means that the principal axis is $C_2$ and the reflection planes are vertical.

Finding the point group of a molecule is a search for these elements. For the molecules of this lecture and its sheet three questions settle it. Which is the principal axis $C_n$? Are there $n$ vertical planes through it? Is there also a plane perpendicular to it, called *horizontal* and written $\sigma_h$? With $n$ vertical planes and nothing more, the group is $C_{nv}$, of order $2n$: water has $n = 2$ and ammonia $n = 3$. A horizontal plane as well brings $n$ twofold axes perpendicular to the principal one, and the group is $D_{nh}$, of order $4n$. Planar $BF_3$ (Sheet 7) has $D_{3h}$. Other combinations give other groups.

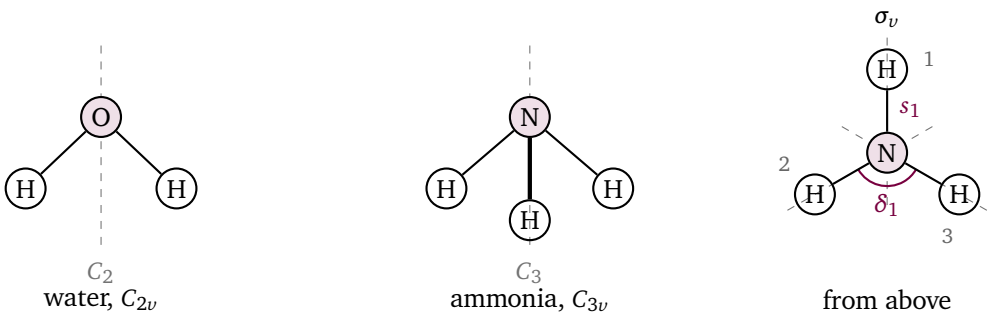



Figure 7.1: Two molecules and their point groups. Ammonia is a pyramid (the thick bond points towards the reader), with the nitrogen on the threefold axis. Seen down that axis (right), each vertical plane holds the nitrogen and one hydrogen. Marked are the stretch $s_1$ of the bond N–$H_1$ and the angle opposite $H_1$, whose change is $\delta_1$ (foreshortened in this view).

The point group of ammonia is the group $C_{3v} \cong S_3$ whose character table Lecture 5 built; that of water is $C_{2v}$, of order 4 (the tutorial).

### The relabelling of nuclei *[self-study]*

**Remark 7.2** (The relabelling the definition suppresses)**.** Number the hydrogens of water $H_1$ and $H_2$. The rotation $C_2$ interchanges them, so it does not leave the *numbered* molecule alone, although the molecule looks exactly the same. The definition asks only that each site be taken by a nucleus of the same species, not that each nucleus return to itself. Like nuclei are indistinguishable (Section 7.2), so we take the operation itself, an element of O(3), as the group element, whatever the numbering. The permutation $\pi(g)$ of the nuclei that $g$ induces still does real work: the atoms $g$ leaves in place are the fixed points of $\pi(g)$, and Lemma 7.3 counts them.

## 7.2 Small oscillations and the dynamical matrix

Lecture 6's block theorem needs an operator that commutes with the group. For a vibrating molecule that operator is the dynamical matrix, and this section builds it.

Consider a molecule of $N$ atoms, labelled $a = 1, \dots, N$, with masses $m_a$. The origin $O$ is the centre of mass (Section 7.1). Every point-group operation fixes it, so the operations act as linear maps. Write $\vec{R}_a^{\,0} \in \mathbb{R}^3$ for the position of atom $a$ when the molecule rests at equilibrium. The superscript 0 is a label meaning "at equilibrium", not a power. Write $\vec{R}_a \in \mathbb{R}^3$ for the position of the same atom at some instant. The vector by which the atom has moved,

$$\vec{u}_a = \vec{R}_a - \vec{R}_a^{\,0}, \qquad \text{equivalently} \qquad \vec{R}_a = \vec{R}_a^{\,0} + \vec{u}_a, \tag{7.1}$$

is the *displacement* of atom $a$ (Figure 7.2). These $N$ small vectors are what the whole lecture is about.

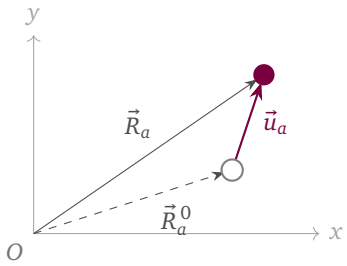



Figure 7.2: One atom: its equilibrium site $\vec{R}_a^{\,0}$ (open circle), its instantaneous position $\vec{R}_a$ (filled) and its displacement $\vec{u}_a = \vec{R}_a - \vec{R}_a^{\,0}$ (greatly exaggerated: the lecture assumes displacements small).

The atoms interact through a potential $V$, a function of the instantaneous positions alone, and each nucleus obeys Newton's law

$$m_a \ddot{\vec{R}}_a = -\nabla_a V, \qquad a = 1, \dots, N,$$

where $\nabla_a$ is the gradient with respect to $\vec{R}_a$ and a dot denotes a time derivative. This is our model of the molecule: point masses held together by the forces $-\nabla_a V$. The light electrons enter only through $V$.

The molecule is isolated, so $V$ depends on those positions only through the interatomic distances $\left|\vec{R}_a - \vec{R}_b\right|$. So $V$ is unchanged by every distance-preserving map of space applied to the whole molecule, reflections included. Nor does $V$ change when two nuclei of the same species exchange places: they are indistinguishable, so $V$ cannot tell which is which. Section 7.4 uses the first invariance, and Section 7.5 uses both.

The relation $\vec{R}_a = \vec{R}_a^{\,0} + \vec{u}_a$ lets us regard $V$ as a function of the displacements. These are small, so $V$ can be expanded in powers of them. We fix the indices as follows. Latin letters $a, b$ label atoms and run over $1, \dots, N$. Greek letters $\alpha, \beta$ label Cartesian directions and run over $x, y, z$. The symbol $u_{a\alpha}$ denotes the $\alpha$-component of $\vec{u}_a$, so that $\vec{u}_a = (u_{ax}, u_{ay}, u_{az})$. A pair $a\alpha$ is thus a *single* index taking $3N$ values, and the displacements of all the atoms stack into one vector of that length,

$$u = \big(\underbrace{u_{1x}, u_{1y}, u_{1z}}_{\vec{u}_1},\ \underbrace{u_{2x}, u_{2y}, u_{2z}}_{\vec{u}_2},\ \dots,\ \underbrace{u_{Nx}, u_{Ny}, u_{Nz}}_{\vec{u}_N}\big) \in \mathbb{R}^{3N}, \tag{7.2}$$

with $u = 0$ meaning the molecule sits exactly at equilibrium. An object carrying two such pairs is a $3N \times 3N$ matrix.

The equilibrium sites $\vec{R}_a^{\,0}$ do not move, so for the displacements Newton's law reads $m_a \ddot{u}_{a\alpha} = -\partial V / \partial u_{a\alpha}$. At equilibrium the forces vanish, so all first derivatives of $V$ are zero at $u = 0$. For small displacements we keep $V$ only up to its quadratic term:

$$V = V(0) + \frac{1}{2} \sum_{a\alpha,\, b\beta} K_{a\alpha,\, b\beta}\, u_{a\alpha} u_{b\beta}, \qquad K_{a\alpha,\, b\beta} = \left. \frac{\partial^2 V}{\partial u_{a\alpha}\, \partial u_{b\beta}} \right|_{u=0}.$$

This is the *harmonic approximation*. In it the force is linear in the displacements, and Newton's law becomes

$$m_a \ddot{u}_{a\alpha} = - \sum_{b\beta} K_{a\alpha,\, b\beta}\, u_{b\beta}.$$

So the molecule is a system of coupled harmonic oscillators, and the $3N \times 3N$ matrix $K$ plays the part of the spring constants. The masses on the left are removed by the *mass-weighted coordinates* $q_{a\alpha} = \sqrt{m_a}\, u_{a\alpha}$, stacked like $u$. Dividing by $\sqrt{m_a}$ and substituting $u_{b\beta} = q_{b\beta} / \sqrt{m_b}$ gives the equation this lecture studies,

$$\ddot{q} = -D\, q, \qquad D_{a\alpha,\, b\beta} = \frac{K_{a\alpha,\, b\beta}}{\sqrt{m_a m_b}}. \tag{7.3}$$

The $3N \times 3N$ matrix $D$ is the *dynamical matrix*. Its rows are indexed by the pair $a\alpha$ and its columns by $b\beta$. The matrix $D$ is real because $V$ is. It is symmetric because mixed partial derivatives commute and the prefactor $1/\sqrt{m_a m_b}$ is symmetric in $a$ and $b$.

From here on $D$ without an argument is the dynamical matrix, not a matrix $D^{(i)}(g)$ of Lecture 6, and $R(g)$ is the $3 \times 3$ orthogonal matrix of a point-group operation, not a position $\vec{R}_a$.

The matrix $D$ is real and symmetric, so $\mathbb{R}^{3N}$ has an orthonormal basis $\varepsilon^{(1)}, \ldots, \varepsilon^{(3N)}$ of eigenvectors of $D$:

$$D\, \varepsilon^{(k)} = \omega_k^2\, \varepsilon^{(k)}, \qquad k = 1, \ldots, 3N.$$

Each eigenvector solves (7.3) by oscillating at a single frequency: with $q(t) = \varepsilon^{(k)} f(t)$ the equation reduces to $\ddot{f} = -\omega_k^2 f$, the single oscillator of earlier courses, so for $\omega_k \neq 0$ the molecule can move as

$$q^{(k)}(t) = A_k\, \varepsilon^{(k)} \cos\bigl(\omega_k t + \varphi_k\bigr), \tag{7.4}$$

with an amplitude $A_k$ and a phase $\varphi_k$. These motions are the *normal modes*: every atom oscillates with the one frequency $\omega_k$, in the fixed pattern $\varepsilon^{(k)}$. The pattern lives in the mass-weighted coordinates: the displacement of atom $a$ is its block of $\varepsilon^{(k)}$ divided by $\sqrt{m_a}$. An eigenvector with $\omega_k = 0$ is a *zero mode*. It does not oscillate at all, and Section 7.4 shows which motions these are.

### Normal modes in full *[self-study]*

The real amplitude $A_k$ and the phase $\varphi_k$ in (7.4) are fixed by the initial conditions. Because the $\varepsilon^{(k)}$ form a basis, the general solution of (7.3) is the superposition

$$q(t) = \sum_{k=1}^{3N} q^{(k)}(t)$$

with one term per mode. For a zero mode the reduced equation is $\ddot{f} = 0$, so $f(t) = A_k + B_k t$, and the motion is uniform rather than oscillatory. This is as it should be for the rigid motions of the molecule (Section 7.4).

Undoing the weighting in full gives the displacement an atom actually undergoes:

$$u_{a\alpha}(t) = \frac{A_k}{\sqrt{m_a}}\,\varepsilon^{(k)}_{a\alpha}\cos\bigl(\omega_k t + \varphi_k\bigr),$$

so a heavy atom moves less than its component of $\varepsilon^{(k)}$ would suggest. In water the oxygen is sixteen times heavier than a hydrogen, so the distinction matters in every mode.

At a stable equilibrium the potential has a minimum at $u = 0$. So $K$, and with it $D$, is a symmetric positive semidefinite matrix: every $\omega_k^2 \geq 0$.

## 7.3 The displacement representation

Next, the point group must act on the space where $D$ acts. That is the $3N$-dimensional real *displacement space* $V_{\text{disp}} = \mathbb{R}^{3N}$, whose vectors we read as displacements $u$, stacked as in (7.2), or, mass-weighted, as $q$. We write $u_a$ for the three-vector block $\vec{u}_a$ sitting at atom $a$, so a displacement is a vector-valued function on the set of atoms, $a \mapsto \vec{u}_a$.

Let $G$ be the point group of the equilibrium configuration. We take it to be *finite*, since the character sums of Lecture 5 and the projectors of Lecture 6 are finite-group tools. Linear molecules, whose nuclei lie on one line and whose full point group is infinite, are therefore set aside. Section 7.4 notes only how their count of vibrations differs.

An operation $g \in G \subset \mathrm{O}(3)$ does two things at once, and keeping them apart is the whole content of the construction. The operation $g$ is an orthogonal map of space, with $3 \times 3$ matrix $R(g)$, so it *rotates* each displacement arrow. It also carries the equilibrium sites among themselves,

$$R(g)\,\vec{R}^{\,0}_a = \vec{R}^{\,0}_{g(a)}, \tag{7.5}$$

that is, it *permutes* the atoms: $g(a)$ is the label of the site that atom $a$ is carried to. This label is well defined. The operation $g$ is a symmetry, so the point $R(g)\vec{R}^{\,0}_a$ is again an equilibrium site, and a nucleus of the same species as atom $a$ sits there. The matrix $R(g)$ is invertible, so it carries distinct sites to distinct sites. The map $a \mapsto g(a)$ is therefore a permutation of the labels $1, \ldots, N$.

For each operation $g$ we define a linear operator $\rho(g)$ on the displacement space that carries the displacement field $u$ along with the molecule. The transported field puts, at the new site, the rotated arrow from the old site:

$$\bigl(\rho(g)\,u\bigr)_{g(a)} = R(g)\,\vec{u}_a, \qquad \text{equivalently} \qquad \bigl(\rho(g)\,u\bigr)_a = R(g)\,\vec{u}_{\,g^{-1}(a)}. \tag{7.6}$$

The two forms say the same thing from opposite ends. The first follows an arrow forward to where it goes. The second asks what has arrived at site $a$ and looks back to $g^{-1}(a)$ to find it. The inverse is the familiar one from the function-space construction of Lecture 3. What is new is that the value is rotated as well as moved. The homomorphism property $\rho(g)\rho(h) = \rho(gh)$ follows as in Lecture 3, now using $R(g)R(h) = R(gh)$ as well. So $\rho$ is a representation of $G$ on $V_{\text{disp}}$, called the *displacement representation*. We write $\chi_{\text{disp}}$ for its character.

The same rule, with the same matrix, holds for the mass-weighted $q$, since $g$ moves each nucleus to one of the same mass. The matrix of $\rho(g)$ is a permutation of $3 \times 3$ blocks, each block the orthogonal $R(g)$, so it is orthogonal.

To count the vibrations, all we need from this representation is its character. The fixed-point lemma of Lecture 5, extended by the $3 \times 3$ trace, gives it at once. Besides rotations and reflections, Lemma 7.3 below covers two more kinds of operation. The *inversion* $i$ through the centre is $\vec{x} \mapsto -\vec{x}$. The *rotoreflection* $S(\theta)$ is a rotation through $\theta$ followed by the reflection in the plane perpendicular to its axis. A reflection is $S(0)$, and the inversion is $S(\pi)$. Chemists write $S_n$ for $S(2\pi/n)$, a subscript that has nothing to do with the symmetric group.

**Lemma 7.3** (Atoms left in place)**.** Let $N_g = \#\{\, a : g(a) = a \,\}$ be the number of atoms left in place by the operation $g$. For a rotation these are the nuclei on its axis, for a reflection those in its plane, and for the inversion or a rotoreflection through a nonzero angle the one at the centre. Then

$$\chi_{\mathrm{disp}}(g) = N_g \cdot \chi_{\mathrm{vec}}(g), \quad \text{where} \quad \chi_{\mathrm{vec}}(g) = \operatorname{tr} R(g) = \begin{cases} 1 + 2\cos\theta, & g = C(\theta), \\ -1 + 2\cos\theta, & g = S(\theta), \end{cases}$$

with $C(\theta)$ a proper rotation through $\theta$ and $S(\theta)$ an improper operation (rotoreflection).

In the matrix of $\rho(g)$, an atom that moves sits off the diagonal and contributes nothing to the trace; an atom left in place contributes $\operatorname{tr} R(g)$. A trace does not depend on the basis, and in a basis with the axis last $C(\theta) = \operatorname{diag}\big(R_{2\times2}(\theta), 1\big)$ and $S(\theta) = \operatorname{diag}\big(R_{2\times2}(\theta), -1\big)$, with $R_{2\times2}(\theta)$ the rotation of the plane perpendicular to the axis. That gives the two traces. The values of $\chi_{\mathrm{vec}}$ needed in practice are

| $g$ | $E$ | $C_2$ | $C_3$ | $C_4$ | $C_6$ | $\sigma = S(0)$ | $i = S(\pi)$ |
|---|---|---|---|---|---|---|---|
| $\chi_{\mathrm{vec}}(g)$ | 3 | −1 | 0 | 1 | 2 | 1 | −3 |

For ammonia (Figure 7.1, middle and right), a rotation $C_3$ leaves only the nitrogen in place, since it alone sits on the axis, and each vertical plane holds the nitrogen and one hydrogen. Section 7.6 completes the count.

*Proof (self-study).* In components, (7.6) reads

$$\big(\rho(g)\, u\big)_{a\alpha} = \sum_{\beta} R(g)_{\alpha\beta}\, u_{g^{-1}(a)\,\beta}.$$

So in the basis labelled by (atom, direction) the matrix of $\rho(g)$ is made of $3 \times 3$ blocks, and the block in row $a$ and column $b$ is $R(g)\,\delta_{a,g(b)}$. Diagonal blocks occur exactly for the atoms with $g(a) = a$, and each contributes $\operatorname{tr} R(g)$. Moved atoms contribute nothing to the trace.

It remains to see that every improper operation is a rotoreflection. By Lecture 2's factorisation $\mathrm{O}(3) = \mathrm{SO}(3) \times \{\pm\mathbb{1}\}$, an improper operation has the form $-\mathbb{1} \cdot Q$ with $Q$ a rotation, through $\varphi$ say. In the basis adapted to the axis of $Q$, the matrix of $-\mathbb{1} \cdot Q$ is

$$\operatorname{diag}\big(-R_{2\times2}(\varphi), -1\big) = \operatorname{diag}\big(R_{2\times2}(\theta), -1\big), \qquad \theta = \varphi + \pi.$$

That is the rotoreflection $S(\theta)$. ∎

## 7.4 Discarding rigid motions

Six of the $3N$ degrees of freedom are not vibrations. A rigid translation, $\vec{u}_a = \vec{t}$ for all $a$, moves no atom relative to any other. An infinitesimal rotation, $\vec{u}_a = \delta\vec{\omega} \times \vec{R}_a^{\,0}$, with $\delta\vec{\omega}$ along

the axis and as long as the small angle turned, is the first-order part of a rigid rotation: each atom moves perpendicular to its position vector (Figure 7.3). It changes the interatomic distances, but only at second order.

Both are zero modes. Write $w$ for either motion, $\vec{w}_a = \vec{t}$ or $\vec{w}_a = \delta\vec{\omega} \times \vec{R}_a^0$, stacked like $u$. The potential $V$ is constant along a rigid motion, so its second derivative along the motion vanishes. At equilibrium that second derivative is $w^{\mathsf{T}} K w$ (for the rotation this takes a short argument, given in self-study below), so $w^{\mathsf{T}} K w = 0$. At a minimum $K$ is positive semidefinite, and then $Kw = 0$: in an orthonormal eigenbasis of $K$, with eigenvalues $\lambda_j \geq 0$ and coordinates $c_j$ of $w$, the sum $w^{\mathsf{T}} K w = \sum_j \lambda_j c_j^2$ vanishes only if every $\lambda_j c_j = 0$.

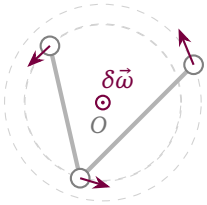
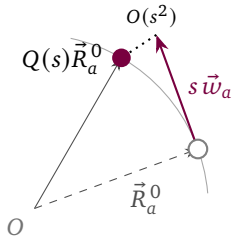



Figure 7.3: An infinitesimal rotation, with $\delta\vec{\omega}$ pointing at the reader: each atom moves perpendicular to its position vector, the more the farther it is from $O$.

*[self-study]*

In detail, take the genuine rigid path $\vec{R}_a(s) = Q(s)\vec{R}_a^0$ with $Q(s)$ a rotation, $Q(0) = \mathbb{1}$. The picture follows one atom: the path is an arc, and the tangent $s\,\vec{w}_a$ departs from it only by $O(s^2)$.




Its displacements are $\vec{u}_a(s) = Q(s)\vec{R}_a^0 - \vec{R}_a^0$, with $u(0) = 0$. Rigidity says that the function

$$f(s) = V\big(u(s)\big)$$

is constant, $f(s) = V(0)$ for every $s$, so $f''(0) = 0$. We compute $f''$ by the chain rule. The first derivative is

$$f'(s) = \sum_{a\alpha} \frac{\partial V}{\partial u_{a\alpha}}\big(u(s)\big)\, \frac{\mathrm{d}u_{a\alpha}}{\mathrm{d}s},$$

and the second is

$$f''(s) = \sum_{a\alpha,\, b\beta} \frac{\partial^2 V}{\partial u_{a\alpha}\, \partial u_{b\beta}}\big(u(s)\big)\, \frac{\mathrm{d}u_{a\alpha}}{\mathrm{d}s}\, \frac{\mathrm{d}u_{b\beta}}{\mathrm{d}s} + \sum_{a\alpha} \frac{\partial V}{\partial u_{a\alpha}}\big(u(s)\big)\, \frac{\mathrm{d}^2 u_{a\alpha}}{\mathrm{d}s^2}.$$

At $s = 0$ the second sum vanishes, because the first derivatives of $V$ vanish at equilibrium. The first sum holds the matrix $K$ and the vector $w = \frac{\mathrm{d}u}{\mathrm{d}s}\big|_0$, so

$$0 = f''(0) = w^{\mathsf{T}} K w.$$

It remains to find $w$. Differentiating $Q(s)^{\mathsf{T}} Q(s) = \mathbb{1}$ at $s = 0$ gives $Q'(0)^{\mathsf{T}} + Q'(0) = 0$, where $Q'(0) = \frac{\mathrm{d}Q}{\mathrm{d}s}\big|_0$, so $Q'(0)$ is antisymmetric. Every antisymmetric $3 \times 3$ matrix acts as a cross product $\vec{x} \mapsto \delta\vec{\omega} \times \vec{x}$ with some vector $\delta\vec{\omega}$. Hence

$$\vec{w}_a = Q'(0)\, \vec{R}_a^0 = \delta\vec{\omega} \times \vec{R}_a^0,$$

as claimed, with $\delta\vec{\omega}$ now the rotation per unit $s$: the displacement $s\,\vec{w}_a$ is the infinitesimal rotation through the small angle $s\,|\delta\vec{\omega}|$. An infinitesimal rotation is a zero mode after all.

We wrote both families of rigid motions, the translations and the rotations, as displacements $u$, while $D$ acts on the mass-weighted coordinates $q$. For $q_{b\beta} = \sqrt{m_b}\,u_{b\beta}$,

$$(Dq)_{a\alpha} = \sum_{b\beta} \frac{K_{a\alpha,\,b\beta}}{\sqrt{m_a m_b}}\,\sqrt{m_b}\,u_{b\beta} = \frac{(Ku)_{a\alpha}}{\sqrt{m_a}},$$

so $Dq = 0$ exactly when $Ku = 0$. In the coordinates $q$ the rigid motions therefore read

$$q_{a\alpha} = \sqrt{m_a}\,t_\alpha \qquad \text{and} \qquad q_{a\alpha} = \sqrt{m_a}\,\big(\delta\vec{\omega} \times \vec{R}_a^{\,0}\big)_\alpha,$$

and they lie in the kernel of $D$. We assume that they span it, that is, that the rigid motions are the *only* zero modes.

A symmetry operation carries rigid motions to rigid motions. So the kernel of $D$ is an invariant subspace of the displacement representation. We define $V_{\text{vib}}$ to be its orthogonal complement in the coordinates $q$. The matrices $\rho(g)$ are orthogonal, so $V_{\text{vib}}$ is invariant too. The genuine vibrations live in $V_{\text{vib}}$, and at the level of characters the separation is a subtraction:

$$\chi_{\text{vib}} = \chi_{\text{disp}} - \chi_{\text{trans}} - \chi_{\text{rot}}.$$

The two subtracted characters are easy. Translations are labelled by the vector $\vec{t}$, which transforms by $R(g)$ itself: $\chi_{\text{trans}} = \chi_{\text{vec}}$. Rotations are labelled by $\delta\vec{\omega}$, which is an *axial* vector, and axial vectors transform with an extra factor, the determinant. Geometrically, a mirror that contains the axis reverses the sense of a rotation about it. Algebraically, by the lemma below and (7.5),

$$R(g)\big(\delta\vec{\omega} \times \vec{R}_a^{\,0}\big) = \big(\det R(g)\, R(g)\,\delta\vec{\omega}\big) \times \vec{R}_{g(a)}^{\,0}.$$

**Lemma 7.4** (Axial vectors). For $R \in \mathrm{O}(3)$, $(R\vec{a}) \times (R\vec{b}) = \det(R)\; R\,(\vec{a} \times \vec{b})$. Consequently $\delta\vec{\omega} \mapsto \det\big(R(g)\big)R(g)\,\delta\vec{\omega}$, and

$$\chi_{\text{rot}}(g) = \det\big(R(g)\big)\,\chi_{\text{vec}}(g) = \begin{cases} 1 + 2\cos\theta, & g \text{ proper}, \\ 1 - 2\cos\theta, & g \text{ improper}. \end{cases}$$

*Proof (self-study).* For proper $R$ the identity is a fact of geometry. Rotations preserve lengths, angles and orientation, and the cross product is defined by exactly those data, so it commutes with rotations. A general orthogonal matrix is $R$ or $-R$ with $R$ proper (Lecture 2). The identity $(-\vec{a}) \times (-\vec{b}) = \vec{a} \times \vec{b}$ supplies the factor $\det(-\mathbb{1}) = -1$ relative to the transformation law of a polar vector. ∎

For a *nonlinear* molecule the rotation labels $\delta\vec{\omega}$ fill three dimensions and

$$\dim V_{\text{vib}} = 3N - 6.$$

A linear molecule has only two independent rotations, since spinning about its own axis moves nothing. So it has $3N - 5$ vibrations, a caveat that matters for $CO_2$.

*[self-study]* In terms of the displacements, orthogonality to the translations in the coordinates $q$ says $\sum_a m_a \vec{u}_a = 0$, so the centre of mass stays put. Orthogonality to the rotations says $\sum_a m_a\,\vec{R}_a^{\,0} \times \vec{u}_a = 0$, so there is no net rotation.

## 7.5 Symmetry block-diagonalises $D$

The group $G$ is a symmetry *of the molecule*, so the potential $V$ does not change when $\rho(g)$ acts on the displacements. To see this, look at where the atoms end up. In the displaced configuration $\rho(g)u$, the atom at site $a$ sits at

$$\vec{R}^{\,0}_a + \big(\rho(g)u\big)_a = R(g)\big(\vec{R}^{\,0}_{g^{-1}(a)} + \vec{u}_{g^{-1}(a)}\big) = R(g)\,\vec{R}_{g^{-1}(a)},$$

by (7.5) and (7.6). This is the old configuration, rotated by $R(g)$, with like nuclei renumbered. Neither the rotation nor the renumbering changes $V$ (Section 7.2). So $V\big(\rho(g)u\big) = V(u)$ for all $g \in G$ and all displacements. For the harmonic potential

$$V = V(0) + \frac{1}{2}\,u^{\mathsf{T}}Ku = V(0) + \frac{1}{2}\,q^{\mathsf{T}}Dq$$

this reads $\rho(g)^{\mathsf{T}} D\,\rho(g) = D$. The matrices $\rho(g)$ are orthogonal, so

$$\big[D,\ \rho(g)\big] = 0 \qquad \text{for all } g \in G. \tag{7.7}$$

So the dynamical matrix plays the part of Lecture 6's Hamiltonian. That lecture's block theorem asks for two things, both now in hand. The group acts on $V_{\text{vib}}$ by orthogonal, hence unitary, matrices (Sections 7.3 and 7.4). And $D$ maps $V_{\text{vib}}$ into itself, because the image of a symmetric matrix is orthogonal to its kernel. On $V_{\text{vib}}$ the matrix $D$ is a real symmetric operator that commutes with every $\rho(g)$. Characters and projectors work over $\mathbb{C}$, so we let the same real matrices act on complex combinations $q_1 + \mathrm{i}q_2$ with $q_1, q_2 \in V_{\text{vib}}$; the physical displacements are the real ones. Decompose

$$V_{\text{vib}} \cong \bigoplus_i m_i\, V^{(i)}.$$

In a symmetry-adapted basis, $D$ acts on each isotypic component through an $m_i \times m_i$ reduced matrix $F^{(i)}$, the same for every partner index. The squared frequencies $\omega^2$ are the eigenvalues of the $F^{(i)}$, each with at least $n_i = \dim V^{(i)}$ independent modes of one symmetry type. A type that occurs once ($m_i = 1$) gives normal modes by symmetry alone. The molecules of this lecture have real characters, so their projectors are real matrices, and the adapted vectors and the modes can be taken real. Group theory fixes the block structure; the spring constants and masses only fill in the small matrices $F^{(i)}$.

*[self-study]* A degeneracy larger than $n_i$ is accidental, or it comes from reality: the displacements are real, so irreducibles with non-real characters pair up (Lecture 6).

The practical recipe, then:

(1) identify the point group $G$ of the equilibrium configuration;
(2) compute $\chi_{\text{disp}}(g) = N_g\,\chi_{\text{vec}}(g)$ class by class;
(3) subtract $\chi_{\text{trans}}$ and $\chi_{\text{rot}}$;
(4) extract multiplicities $m_i = \langle \chi_i, \chi_{\text{vib}} \rangle_G$ from the character table;
(5) construct symmetry-adapted coordinates with the projectors of Lecture 6, and diagonalise each reduced $F^{(i)}$ that is larger than $1 \times 1$.

## 7.6 Worked example: ammonia

Ammonia is a triangular pyramid of $N = 4$ atoms. We find its point group by answering the three questions of Section 7.1. The principal axis is a $C_3$ through the nitrogen. Three

vertical planes pass through it, each through the nitrogen and one hydrogen. There is no horizontal plane, since the nitrogen lies off the plane of the hydrogens. So the group is $C_{3v}$. Its character table was built in Lecture 5. The classes have sizes $1, 2, 3$, and the character bookkeeping is

| | $E$ | $2C_3$ | $3\sigma_v$ |
|---|---|---|---|
| $N_g$ | 4 | 1 | 2 |
| $\chi_{\text{vec}}$ | 3 | 0 | 1 |
| $\chi_{\text{disp}}$ | 12 | 0 | 2 |
| $\chi_{\text{trans}}$ | 3 | 0 | 1 |
| $\chi_{\text{rot}}$ | 3 | 0 | $-1$ |
| $\chi_{\text{vib}}$ | 6 | 0 | 2 |
| $\chi_{A_1}$ | 1 | 1 | 1 |
| $\chi_{A_2}$ | 1 | 1 | $-1$ |
| $\chi_E$ | 2 | $-1$ | 0 |

It remains to find how often each irreducible representation occurs in $V_{\text{vib}}$. In the multiplicity formula $m_i = \langle \chi_i, \chi_{\text{vib}} \rangle_G$ each class contributes its size times the two characters, and the sum is divided by the order $|C_{3v}| = 6$:

$$m_{A_1} = \frac{1}{6}\big[1 \cdot 1 \cdot 6 + 2 \cdot 1 \cdot 0 + 3 \cdot 1 \cdot 2\big] = 2,$$
$$m_{A_2} = \frac{1}{6}\big[6 + 0 - 6\big] = 0,$$
$$m_E = \frac{1}{6}\big[1 \cdot 2 \cdot 6 + 2 \cdot (-1) \cdot 0 + 3 \cdot 0 \cdot 2\big] = 2.$$

Chemists write $\Gamma_X$ for the representation with character $\chi_X$ and list it by its irreducible types:

$$\Gamma_{\text{vib}} = 2A_1 \oplus 2E.$$

These are six vibrational dimensions ($3N - 6 = 6$ ✓). Generically, that is barring the accidents that Lecture 6's corollary on symmetry-enforced degeneracy leaves open, there are four distinct frequencies: two nondegenerate $A_1$ frequencies and two doubly degenerate $E$ frequencies. The double degeneracies are not accidents of a particular model of the forces. They are theorems.

For explicit modes, internal coordinates serve best (Figure 7.1, right): the three bond stretches $s_k$, the increase in length of the bond N–H$_k$, and the three angle changes $\delta_k$, the increase of the H–N–H angle opposite H$_k$. These are six coordinates for six vibrations. Each is, to first order, a linear function of $q$, and we use the same letter for its vector: $s_k(q) = s_k^{\mathsf{T}} q$, and likewise for $\delta_k$. An operation $g$ that takes H$_k$ to H$_{g(k)}$ carries bond $k$ to bond $g(k)$, and the angle opposite H$_k$ to the angle opposite H$_{g(k)}$, so

$$\rho(g)\, s_k = s_{g(k)}, \qquad \rho(g)\, \delta_k = \delta_{g(k)}.$$

So the group permutes the six vectors, and they are what we project. For ammonia they form a basis of $V_{\text{vib}}$ (self-study below).

*[self-study]* Here we used that none of the coordinates changes under a rigid motion, so each vector is orthogonal to the rigid motions and lies in $V_{\text{vib}}$; for ammonia the six are independent.

Moreover, the displacement $\rho(g)q$ stretches bond $g(k)$ as $q$ stretches bond $k$, so $s_{g(k)}^{\mathsf{T}}\rho(g)q = s_k^{\mathsf{T}}q$ for every $q$. Hence $\rho(g)^{\mathsf{T}}s_{g(k)} = s_k$, and $\rho(g)$ is orthogonal, so $\rho(g)s_k = s_{g(k)}$.

The rotations cycle the indices, and the reflection in the plane through $H_1$ swaps $2 \leftrightarrow 3$, for the $s_k$ and the $\delta_k$ alike. Each triple therefore carries the permutation representation of $S_3$, which decomposes as $A_1 \oplus E$ (Lecture 6's example of the projectors on the triangle's orbitals), and

$$\Gamma_{\text{vib}} = (A_1 \oplus E)_{\text{stretch}} \oplus (A_1 \oplus E)_{\text{bend}},$$

in agreement with the Cartesian count. The projectors are *exactly* those of that example: $P_{A_1} = J/3$ and $P_E = \mathbb{1} - J/3$ on each triple, with $J$ the $3 \times 3$ matrix of ones. Applied to $s_1$ and to $\delta_1$, the projector $P_{A_1}$ gives the two $A_1$ coordinates, up to a factor $\frac{1}{3}$:

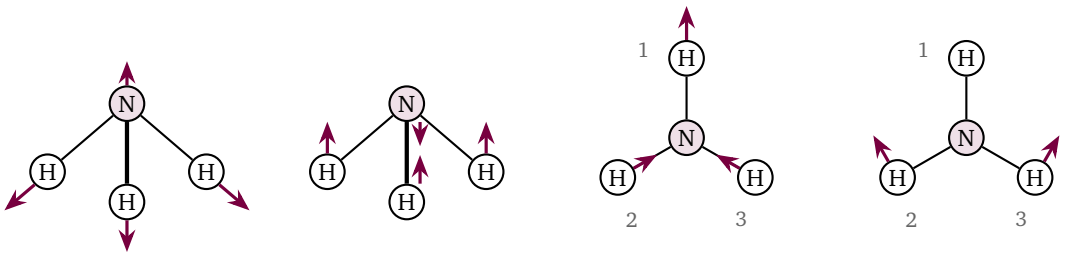
$$\underbrace{s_1 + s_2 + s_3}_{\text{symmetric stretch } (A_1)}, \qquad \underbrace{\delta_1 + \delta_2 + \delta_3}_{\text{umbrella } (A_1)}.$$

In the symmetric stretch the three bonds lengthen together. In the umbrella the three angles open together: the pyramid flattens, and the nitrogen moves along the axis against the hydrogens (Figure 7.4).

From each triple $P_E$ gives a pair of $E$ partners. For the stretches,

$$P_E\, s_1 = \frac{1}{3}\,(2s_1 - s_2 - s_3), \qquad P_E\,(s_2 - s_3) = s_2 - s_3,$$

the asymmetric-stretch pair, and the same with $\delta_k$ in place of $s_k$ gives the asymmetric-bend pair. Symmetry alone does not make these normal modes. The two $A_1$ coordinates generically mix through a $2\times2$ reduced matrix. So do the two $E$ pairs, with one $2\times2$ matrix for both partners, because the two pairs are built alike. With the six zero modes set aside, a $12 \times 12$ eigenvalue problem has become two $2 \times 2$ problems.




(a) sym. stretch ($A_1$) (b) umbrella ($A_1$) (c) asym. stretch ($E$) (d) asym. bend ($E$)

Figure 7.4: Ammonia's symmetry coordinates. Seen from the side: (a) the symmetric stretch $s_1 + s_2 + s_3$ and (b) the umbrella $\delta_1 + \delta_2 + \delta_3$. Seen down the axis: the first partners (c) $2s_1 - s_2 - s_3$ and (d) $2\delta_1 - \delta_2 - \delta_3$ of the two $E$ pairs. The arrows are schematic, and in the top views the small recoil of the nitrogen is not drawn. Coordinates of one type mix, so these are symmetry coordinates, not yet normal modes.

The measured spectrum agrees. Ammonia shows four fundamental frequencies, absorption lines at the mode frequencies $\omega_k$ themselves rather than at their sums or multiples. They

are quoted as wavenumbers $\tilde{\nu} = \omega/2\pi c$, with $c$ the speed of light (NIST, after Shimanouchi):

| mode | type | $\tilde{\nu}$ ($\mathrm{cm}^{-1}$) |
|---|---|---|
| symmetric stretch | $A_1$ | 3337 |
| umbrella | $A_1$ | 950 |
| asymmetric stretch | $E$ | 3444 |
| asymmetric bend | $E$ | 1627 |

Modes of one type mix, so each name records the internal coordinate that dominates the mode. Which of these vibrations infrared light can excite is a further question, answered in Lecture 8.

*[self-study]* The umbrella line is in fact split in two, because the nitrogen can tunnel through the plane of the hydrogens. That effect lies outside the harmonic picture.

*[tutorial]* The $E$ pairs. Show that $P_E\, s_2 - P_E\, s_3 = s_2 - s_3$, so that $(2s_1 - s_2 - s_3,\ s_2 - s_3)$ are two orthogonal vectors of the same $E$ component: even and odd under the reflection through $\mathrm{H}_1$, like $x$ and $y$. The bend partners $(2\delta_1 - \delta_2 - \delta_3,\ \delta_2 - \delta_3)$ come out the same way. Check that $C_3$ and this reflection transform both pairs by the same $2 \times 2$ matrices.

*[self-study]* *Internal coordinates: a caveat.* For water and ammonia the natural internal coordinates are exactly $3N - 6$ independent ones, so the bookkeeping closes. In general they must be checked on two counts. They can be *redundant*: around a ring, or around an atom with many neighbours, the angles obey relations. And they can be *incomplete*. In planar $\mathrm{BF}_3$ (Sheet 7) the three stretches and the three bond angles carry $2A_1' \oplus 2E'$, whereas the vibrations are $A_1' \oplus 2E' \oplus A_2''$ (one prime marks a type even under $\sigma_h$, two primes a type odd under it). The three angles always add up to $2\pi$, which costs an $A_1'$, and no bond length or angle sees the umbrella motion out of the plane, which is the $A_2''$. The Cartesian route of Sections 7.3–7.4 is never ambiguous.

## 7.7 Worked example: water *[tutorial]*

Water is bent: $N = 3$, point group $C_{2v} = \{E, C_2, \sigma_v, \sigma_v'\}$ (Section 7.1): the half-turn $C_2$ about the axis that bisects the H–O–H angle, and the reflections in two planes through that axis. Every operation squares to $E$, and composing two distinct nonidentity operations gives the third. So $C_{2v}$ is not cyclic: it is the Klein four-group named in Lecture 1. We place the molecule in the $yz$-plane with the $C_2$ axis along $z$. Let $\sigma_v$ denote the molecular plane ($yz$) and $\sigma_v'$ the perpendicular plane ($xz$). The group is abelian of order 4, so its four irreducible characters are one-dimensional sign characters. With Mulliken labels they are

| $C_{2v}$ | $E$ | $C_2$ | $\sigma_v(yz)$ | $\sigma_v'(xz)$ | |
|---|---|---|---|---|---|
| $A_1$ | 1 | 1 | 1 | 1 | $z$ |
| $A_2$ | 1 | 1 | −1 | −1 | |
| $B_1$ | 1 | −1 | 1 | −1 | $y$ |
| $B_2$ | 1 | −1 | −1 | 1 | $x$ |

The last column records how the coordinate functions transform, which we need shortly. Note that interchanging the two planes interchanges the labels $B_1 \leftrightarrow B_2$. Conventions differ between books, so fix one and stay consistent. Most tables, NIST's among them, also put water in the $yz$-plane but attach $B_1$ to the other plane, $xz$. There water's antisymmetric stretch is $B_2$; here it is $B_1$.

*Step 1: characters.* The numbers of atoms left in place and the traces are

| | $E$ | $C_2$ | $\sigma_v(yz)$ | $\sigma'_v(xz)$ |
|---|---|---|---|---|
| $N_g$ | 3 | 1 | 3 | 1 |
| $\chi_{\text{vec}}$ | 3 | −1 | 1 | 1 |
| $\chi_{\text{disp}} = N_g\chi_{\text{vec}}$ | 9 | −1 | 3 | 1 |
| $\chi_{\text{trans}}$ | 3 | −1 | 1 | 1 |
| $\chi_{\text{rot}} = \det\cdot\chi_{\text{vec}}$ | 3 | −1 | −1 | −1 |
| $\chi_{\text{vib}}$ | 3 | 1 | 3 | 1 |

Here $N_{C_2} = 1$ because only the oxygen sits on the axis, $N_{\sigma_v} = 3$ because the whole molecule lies in its own plane, and $N_{\sigma'_v} = 1$ because only the oxygen lies in the perpendicular plane.

*Step 2: multiplicities.* All class sizes are 1 and $|G| = 4$, so

$$m_{A_1} = \frac{1}{4}(3+1+3+1) = 2, \qquad m_{A_2} = \frac{1}{4}(3+1-3-1) = 0,$$
$$m_{B_1} = \frac{1}{4}(3-1+3-1) = 1, \qquad m_{B_2} = \frac{1}{4}(3-1-3+1) = 0,$$

so

$$\Gamma_{\text{vib}} = 2A_1 \oplus B_1,$$

that is, three vibrational coordinates ($3N - 6 = 3$ ✓). *Two* of them have the full symmetry $A_1$, and one is antisymmetric under both $C_2$ and $\sigma'_v$. Generically the $2\times 2$ reduced $A_1$ block has two distinct eigenvalues, neither equal to the $B_1$ one, so all three frequencies are distinct. Any coincidence, inside the block or across the blocks, would be accidental.

*Step 3: symmetry-adapted coordinates by projection.* What the projectors return is a basis adapted to the symmetry, not eigenvectors of $D$. Symmetry guarantees that a projected vector is a normal mode only when its symmetry type occurs once. Where an irreducible occurs $m_i > 1$ times, the reduced matrix $F^{(i)}$ generically mixes the copies, and the modes follow only after $F^{(i)}$ is diagonalised.

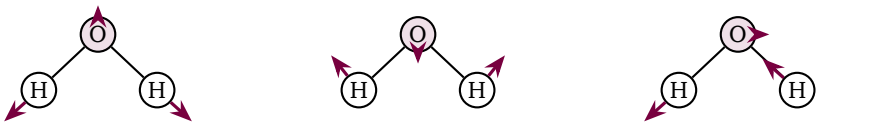
As for ammonia, the vibrational physics is most transparent in internal coordinates. For water they are the two bond stretches $s_1, s_2$ (elongations of the O–H bonds) and the bend $\delta$ (the increase of the H–O–H angle): three coordinates for three vibrations. The group permutes them simply. The operations $C_2$ and $\sigma'_v$ swap $s_1 \leftrightarrow s_2$, while $E$ and $\sigma_v$ fix each. The bend $\delta$ is invariant under everything, so it is pure $A_1$. Applying the projectors of Lecture 6 to the trial vector $s_1$ gives

$$P_{A_1}s_1 = \frac{1}{4}\big(s_1 + s_2 + s_1 + s_2\big) = \frac{1}{2}(s_1 + s_2),$$
$$P_{B_1}s_1 = \frac{1}{4}\big(s_1 + (-1)s_2 + s_1 + (-1)s_2\big) = \frac{1}{2}(s_1 - s_2),$$

with $P_{A_2}s_1 = P_{B_2}s_1 = 0$. So the three symmetry-adapted vibrations of water are (Figure 7.5)

$$\underbrace{s_1 + s_2}_{\text{symmetric stretch } (A_1)}, \qquad \underbrace{\delta}_{\text{bend } (A_1)}, \qquad \underbrace{s_1 - s_2}_{\text{antisymmetric stretch } (B_1)}.$$

Lecture 6's block theorem adds a final refinement that pictures of modes do not show. The two $A_1$ coordinates span a single isotypic component with $m_{A_1} = 2$. So symmetry does *not* make the symmetric stretch and the bend normal modes. Generically the actual




$\nu_1$: sym. stretch ($A_1$) $\nu_2$: bend ($A_1$) $\nu_3$: antisym. stretch ($B_1$)

Figure 7.5: The three symmetry-adapted vibrations of water with their Mulliken labels. The arrows are schematic. Those on the oxygen show the direction of its recoil, which keeps the centre of mass at rest; to scale they would be far shorter, since the oxygen is sixteen times heavier than a hydrogen. The two $A_1$ coordinates generically mix, and their $2 \times 2$ reduced matrix gives the two $A_1$ frequencies.

eigenvectors are two combinations of them, orthogonal in the mass-weighted coordinates $q$ and determined by the $2 \times 2$ reduced matrix $F^{(A_1)}$, whose entries depend on the force constants. The $B_1$ mode is alone in its symmetry type, so it is an exact normal mode by symmetry alone. Group theory has reduced a $9 \times 9$ problem to a $2 \times 2$ problem ($A_1$), a $1 \times 1$ problem ($B_1$), and the six rigid modes already known to have zero frequency.

**Summary.**

- *Point groups.* A molecule's symmetry operations form its point group; ammonia's is $C_{3v} \cong S_3$, water's $C_{2v}$.
- *Vibrations as a representation.* Nuclear displacements carry a (generally reducible) representation of the molecule's point group (eq. (7.6)).
- *Mode counting.* The displacement character is read off the atoms a symmetry leaves in place (Lemma 7.3). Removing the translations and rotations (Lemma 7.4) leaves the vibrational content, $3N - 6$ dimensions for a nonlinear molecule.
- *Symmetry commutes.* The point group commutes with the dynamical matrix (eq. (7.7)), so Lecture 6's block theorem applies: frequencies are labelled by irreducibles, each at least $\dim V^{(i)}$-fold degenerate.
- *Ammonia.* $\Gamma_{\text{vib}} = 2A_1 \oplus 2E$: generically four frequencies, two of them doubly degenerate, before any force constant is known, and four are measured. The projectors of Lecture 6 give the symmetry-adapted coordinates; symmetry guarantees that they are normal modes only where a type occurs once.

## References

Bishop, *Group Theory and Chemistry*; Tinkham, *Group Theory and Quantum Mechanics*, §§7-11–7-13 (molecular vibrations); Elliott–Dawber, *Symmetry in Physics*, Ch. 6 (vibrational analysis).